

ICT Department

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ITLM 801 MACHINE_ LEARNING

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Heart Disease Risk Prediction System report

Introduction

Heart disease is a major public health challenge that places a significant burden on healthcare systems. Early detection and risk stratification are essential for improving patient outcomes and reducing treatment costs. Traditional diagnosis relies heavily on manual assessment by clinicians, which can be time-consuming and subject to human error.

With the advancement of machine learning and data-driven technologies, intelligent clinical decision support systems can assist healthcare professionals by providing accurate and consistent predictions based on patient data. This project focuses on developing a machine learning-based heart disease risk prediction system and deploying it as a web application for real-time use in a hospital environment.

At CHUB referral and teaching hospital, clinicians manage a large number of patients with cardiovascular conditions daily. The manual evaluation of patient risk levels using clinical reports and test results can be slow, inconsistent, and prone to subjective judgment. This can lead to delayed diagnosis, inefficient resource allocation, and increased risk of severe complications for high-risk patients.

Therefore, there is a need for an automated, reliable, and easy-to-use system that can:

1. Quickly assess a patient's heart disease risk level
2. Support clinicians in decision-making
3. Improve early detection and timely intervention

Objectives

- To design and develop an intelligent heart disease risk prediction system using machine learning and deploy it as a web-based clinical decision support tool.
 - To collect and explore a structured heart disease dataset with clinical and demographic features
 - To perform data preprocessing, including handling missing values, scaling, and encoding
 - To train and evaluate multiple machine learning models for multi-class classification
 - To select the best-performing model based on accuracy and evaluation metrics
 - To develop a user-friendly web interface using Flask
 - To deploy the trained model for real-time patient risk prediction
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Dataset Description

The dataset consists of 5,000 patient records. Each patient is described by 12 clinical and demographic input features and a five-class target label.

Input Features

- Age
- Resting Blood Pressure (trestbps)
- Serum Cholesterol (chol)
- Maximum Heart Rate Achieved (thalach)
- ST Depression (oldpeak)

- Sex
- Chest Pain Type (cp)
- Thalassemia Status (thal)
- Resting ECG Results (restecg)
- Slope of Peak Exercise ST Segment (slope)
- Exercise Induced Angina (exang)
- Number of Major Vessels Colored (ca)
- Fasting Blood Sugar (fbs)

Target Variable

Heart disease risk category:

- 0 = No disease
- 1 = Very mild
- 2 = Mild
- 3 = Severe
- 4 = Immediate danger

System Architecture

The system follows a modular architecture consisting of four main layers:

1. Data Layer – Stores the dataset and trained machine learning model
2. Processing Layer – Performs preprocessing, feature transformation, and prediction
3. Application Layer – Flask backend that handles user requests and model inference
4. Presentation Layer – Web-based user interface for data input and result display

Data Preprocessing

- Handling missing values using SimpleImputer
- Encoding categorical variables using LabelEncoder or One-Hot Encoding

- Feature scaling using StandardScaler
- Splitting the dataset into training and testing sets

Exploratory Data Analysis (EDA)

- Statistical summary of features
- Visualization of class distribution
- Correlation analysis between features
- Detection of missing values and outliers

Model Training

Several classification models were trained and evaluated, including:

- Logistic Regression
- Random Forest Classifier
- Support Vector Machine (SVM)
- Gradient Boosting

Model Evaluation

- Accuracy score
- Confusion matrix
- Precision, Recall, and F1-score
- Cross-validation

The best-performing model was 'Random Forest: 0.999 =99%

Backend (Flask)

The Flask framework was used to:

- Load the trained model and preprocessing pipeline
- Accept user input from the web form
- Convert input data into a format suitable for the model
- Generate and return prediction results

Frontend

- HTML forms for entering patient details

- CSS for layout and styling
- A results page displaying the predicted risk level

Testing and Results

The system was tested using both training data samples and manually entered patient data through the web interface. The model demonstrated reliable classification performance across all five risk categories. Predictions were generated in real time, and the web interface responded within seconds, making it suitable for clinical use.

Conclusion

This project successfully developed an end-to-end heart disease risk prediction system using machine learning and Flask deployment. The system provides a fast, accurate, and user-friendly platform for assisting healthcare professionals in identifying high-risk patients. With further enhancements and real-world integration, the system can contribute significantly to improving early diagnosis and patient outcomes.